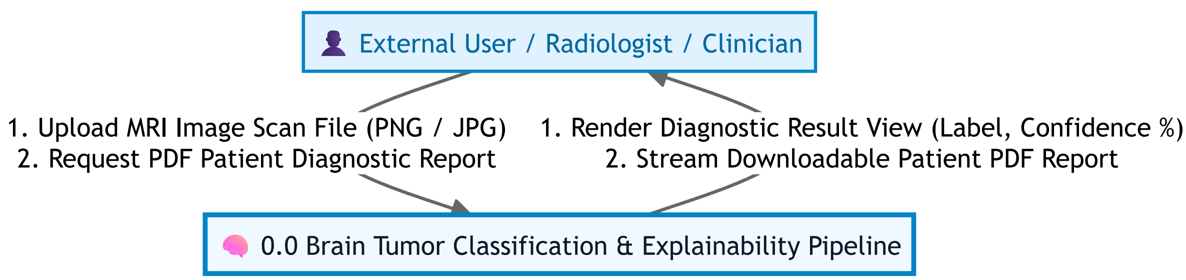
🧠 Brain Tumor Detection System — All Project Diagrams

Complete Visual Architecture, Data Flow, E-R, Class, Object Interaction, Control Flow, and Sequence Diagrams for RK University Project Report

💡 **Screenshot Tip:** Scroll to any diagram below, press `Win + Shift + S` to capture a high-resolution screenshot, and paste it directly into your Project Report Word document!

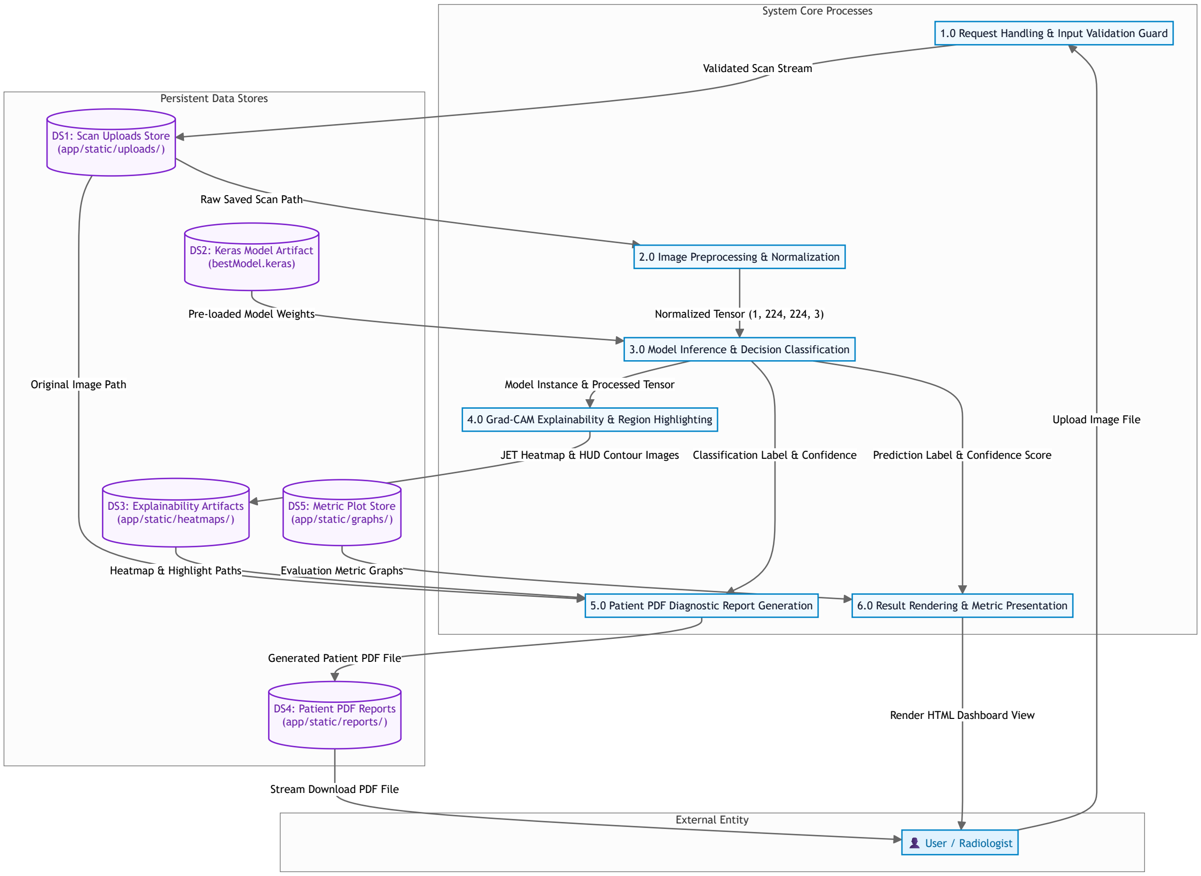
1. Figure 4.1 (a): Data Flow Diagram (DFD Level 0 — Context Diagram)

Context-level boundary mapping user inputs and system output deliverables.



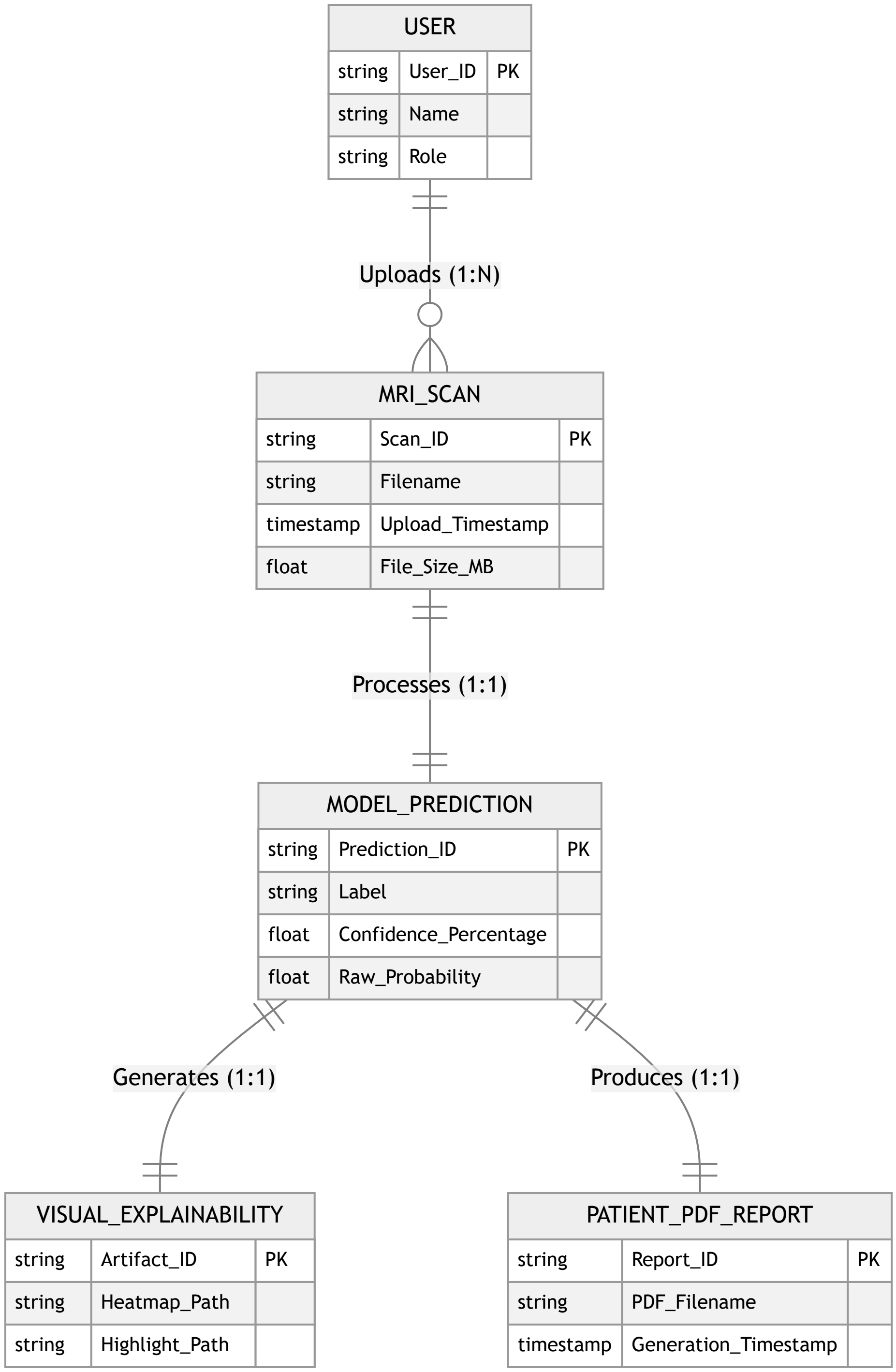
2. Figure 4.1 (b): Data Flow Diagram (DFD Level 1 — Core Pipeline & Data Stores)

System decomposition into 6 core processes and 5 persistent data stores.



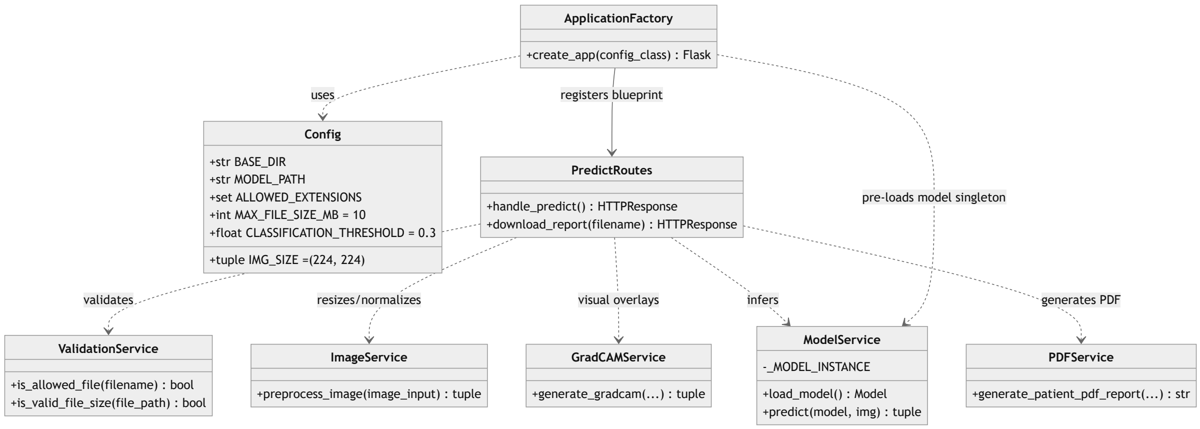
3. Entity-Relationship (E-R) Diagram

Entity-relationship model of system users, MRI scan files, deep learning predictions, visual overlays, and patient PDF reports.



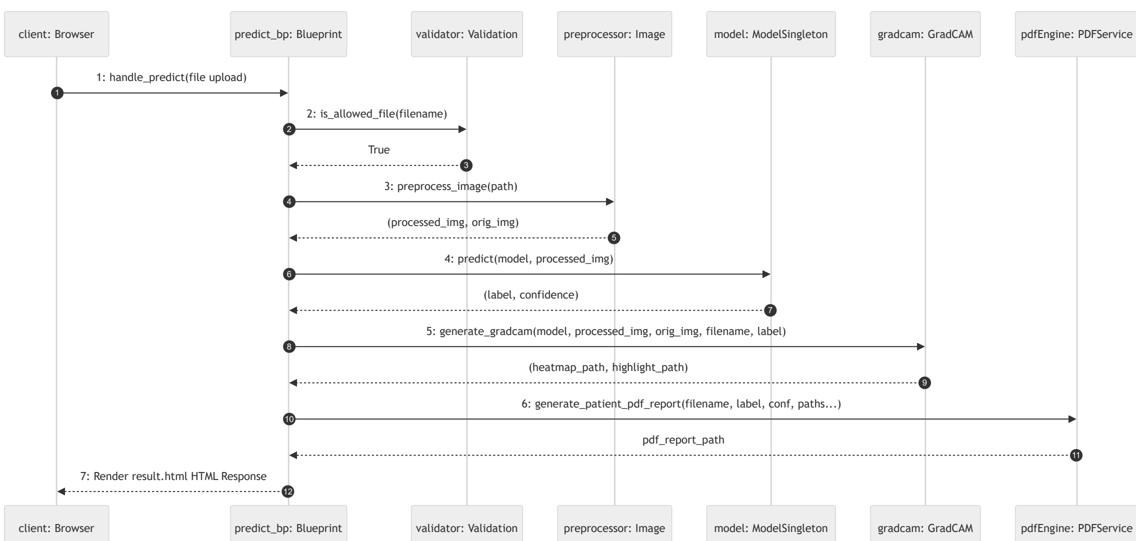
4. Figure 4.2: High-Level System Class Architecture Diagram

UML Class Architecture depicting Config, Application Factory, Controller Blueprints, and Service Layer Singletons.



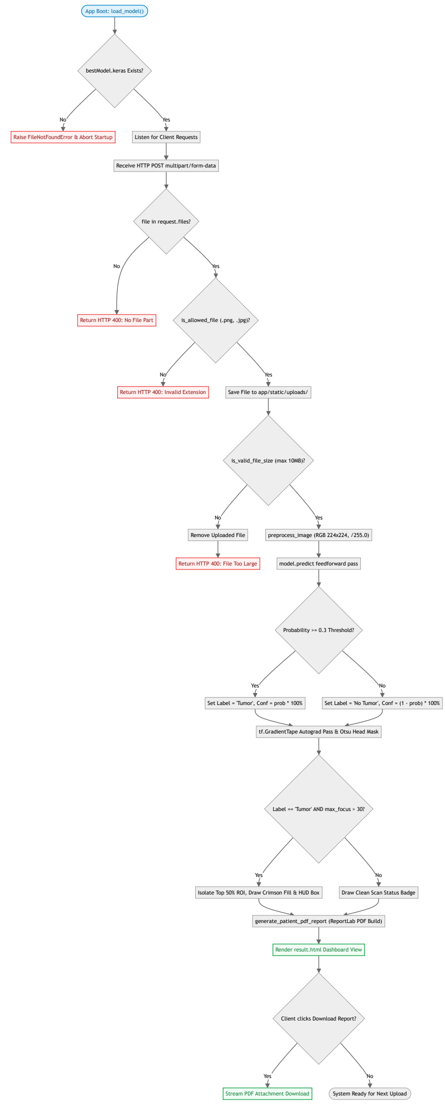
5. Section 4.7.5: Object Interaction Diagram

UML Object Interaction Diagram depicting dynamic method calls between instantiated runtime service objects.



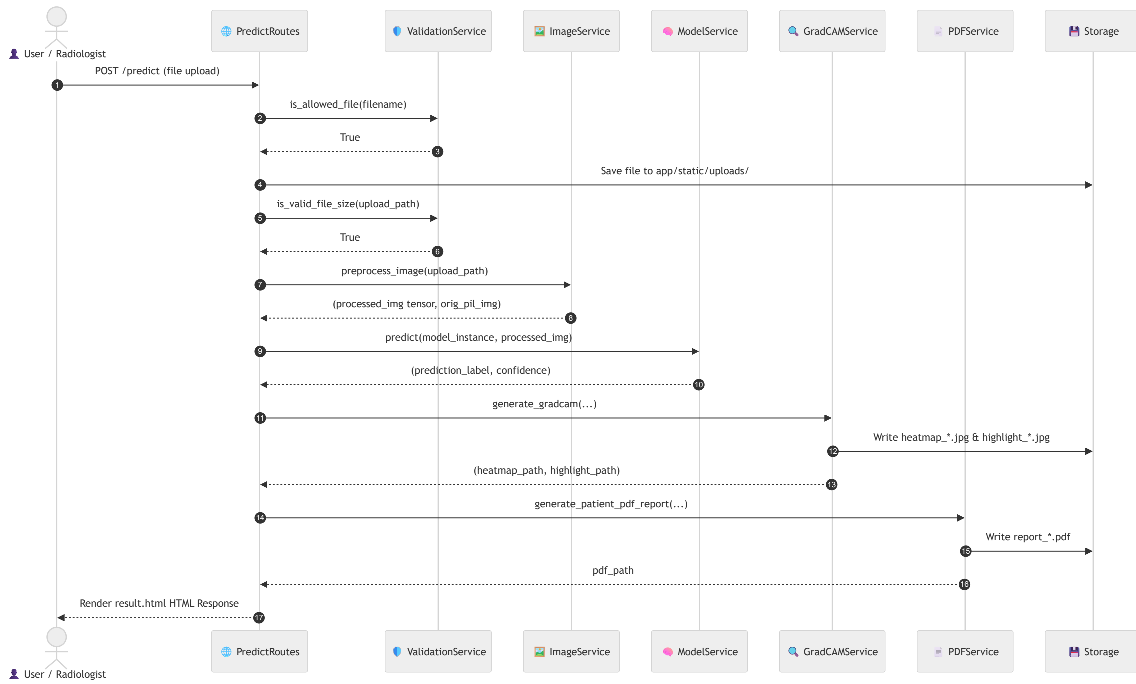
6. Section 4.8.3: Control Flow Diagram (CFD)

Control Flow Diagram mapping execution states, decision predicates, exception branches, and threshold branching.



7. Component Interaction & Execution Sequence Diagram

Runtime sequence of service layer invocations for synchronous HTTP requests.



User / Radiologist
PredictRoutes
ValidationService
ImageService
ModelService
GradCAMService
PDFService
Storage
POST /predict (file upload)
is_allowed_file(filename)
True
Save file to app/static/uploads/
is_valid_file_size(upload_path)
True
preprocess_image(upload_path)
(processed_img tensor, orig_pil_img)
predict(model_instance, processed_img)
(prediction_label, confidence)
generate_gradcam(...)
Write heatmap_*.jpg & highlight_*.jpg
(heatmap_path, highlight_path)
generate_patient_pdf_report(...)
Write report_*.pdf
pdf_path
Render result.html HTML Response